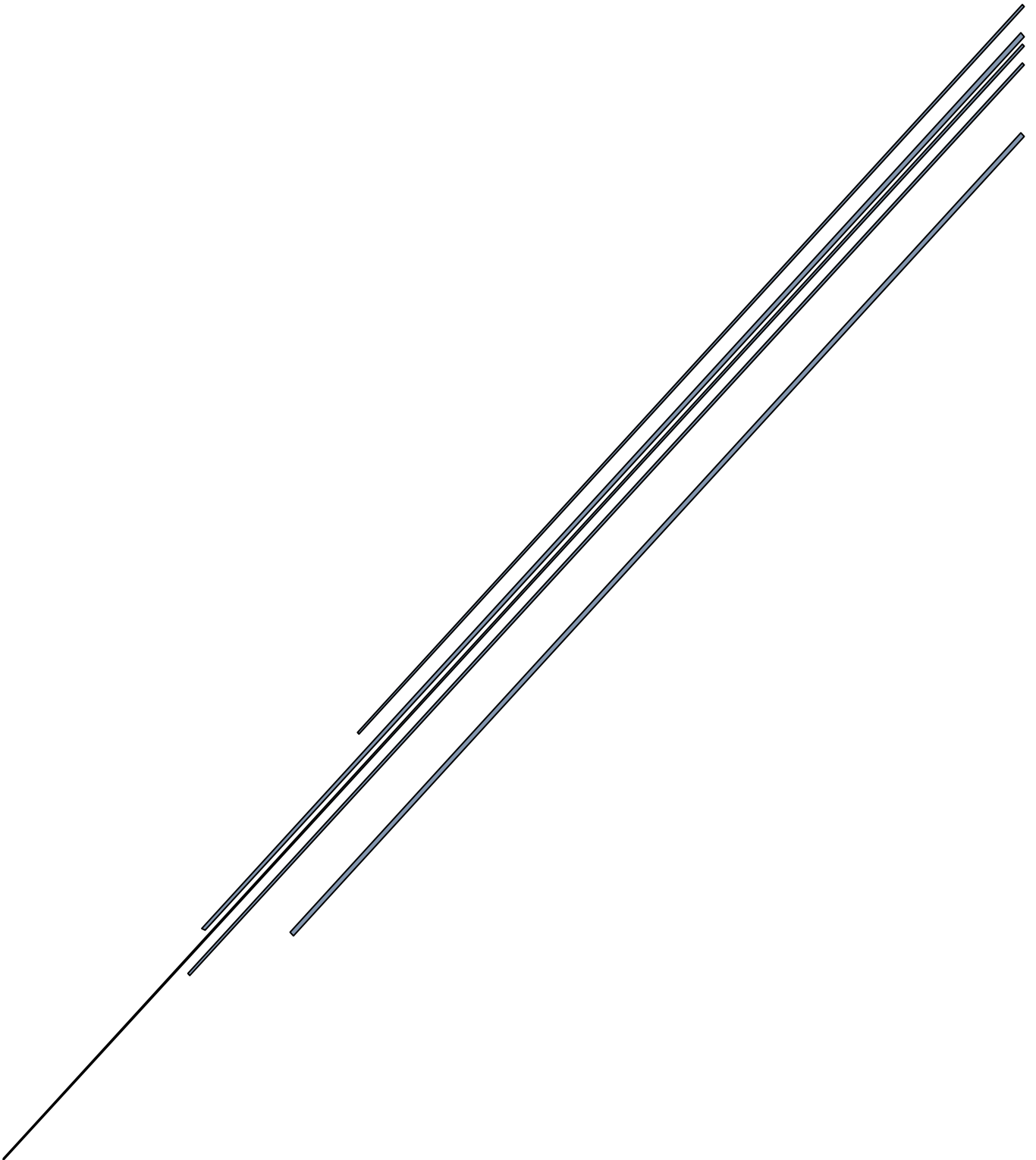


COS 341 SEMESTER PROJECT

User Manual

Aren Repko (u04515791) and Daniel Geerdink (u22556860)



Assumptions

- Each token is separated by a space (parentheses, commas, and semicolons cannot follow directly after another token), a space needs to be between them.
- You may have endlines and tabs
- Please see next page for an example input

Output

- At each stage a file is generated for each output:
- **Tokens.xml** for the Token Stream or result of the Lexer (Part 1)
- **SyntaxTree.xml** for the Syntax tree or the result of the Parser (Part 2)
- **SymbolTable.txt** for the VTable or Symbol table (Part 3)
- Result of the type checking is printed in the console (Part 4)
- **intermediateCode.txt** has the intermediate code (Part 5a)

How to run

- A single .jar file is given: **Compiler.jar**
- Place the input file in the same directory as the .jar file
- Use this command in the terminal to run the executable: **java -jar Compiler.jar**
- The output files will be added to the directory of the executable

Completed Parts

- Part 1 to Part 5a was completed (Up to intermediate code generation)

Example code format:

```
main
num V_a , text V_b , num V_c , num V_d ,
begin
    V_a = 6.7734 ;
    V_b = "Test" ;
    V_c = 8 ;

    V_d = F_calculate ( V_a , 9.32 , V_c ) ;
    V_d < input ;

    if and ( eq ( V_a , V_c ) , grt ( V_a , 40.1 ) ) then
        begin
            print V_b ;
        end
    else
        begin
            if eq ( V_a , V_c ) then
                begin
                    V_a = sqrt ( 99 ) ;
                    V_c = add ( 0 , 450.1 ) ;
                    V_a = mul ( 3 , V_c ) ;
                    V_c = div ( V_a , 8 ) ;
                    V_c = sub ( V_a , V_c ) ;
                end
            else
                begin
                    end ;
                end ;
        end ;
    end ;

end

num F_calculate ( V_a , V_b , V_c ) {
    num V_t1 , text V_t2 , num V_t3 ,
    begin
        V_t1 = add ( V_a , V_b ) ;
        if grt ( V_c , V_t1 ) then
            begin
                return V_c ;
            end
        else
            begin
                V_a = F_calculate ( 8 , 10 , V_c ) ;
            end ;
        end
    end
}

end
```